

NITROGEN AND SULPHUR

Photochemical Smog

Photo means light and smog is a mixture of smoke and fog.

“Photochemical smog is a mixture of pollutants that are formed when oxides of nitrogen and volatile organic compounds (VOCs) react with sunlight, creating a brown haze.”

Causes

Nitrogen oxides (NO_x), including nitric oxide (NO) and nitrogen dioxide (NO₂), the causes.

Reactions of NO and NO₂ with unburned hydrocarbons, leading to the formation of peroxyacetyl nitrate (PAN).

Peroxyacetyl nitrate (PAN) which is secondary pollutant, an eye irritant is produced by the photochemical smog.

In the morning hours, the NO and hydrocarbon level increases followed by an increase in nitrogen dioxide.



Mechanism of the Formation of PAN

1. Formation of Peroxyacetyl Radicals:

Unburned hydrocarbons, such as volatile organic compounds (VOCs) are present in the atmosphere.

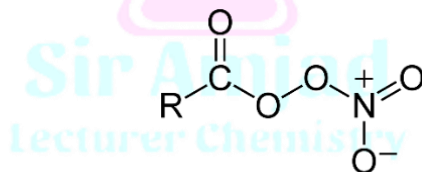
In the presence of sunlight, NO₂ can undergo photo dissociation, splitting into NO and an oxygen atom (O).

The oxygen atom reacts with the unburned hydrocarbons, forming peroxyacetyl radicals (RCO₃).

Oxygen + Unburned HC → peroxyacetyl radicals

2. Reaction with NO

The peroxyacetyl radicals (RCO₃) formed reacts with NO leading to the formation of peroxyacetyl nitrate (PAN).



3. Problem associated with PAN

The formation of PAN is problematic because it is a stable compound that can be transported over long distances.

It acts as a reservoir for NO_x, preventing their conversion back to NO and NO₂, which are important in controlling air pollution.

Moreover, PAN can undergo thermal decomposition in the lower atmosphere, releasing NO₂ and reactive organic radicals, which further contribute to the formation of smog and photochemical reactions.

Uses of Sulphur and its Compounds

1. Fertilizer

- Sulphur compounds like Sulphuric acid and ammonium sulphate are used in fertiliser to provide sulphur to crops.
- Sulphur helps in the synthesis of important amino acids and vitamins in plants,

2. Gunpowder

- Gunpowder typically consists of sulphur, charcoal and potassium nitrate.

- Sulphur plays an important role in the combustion and explosive properties of gunpowder

3. Rubber Industry

Sulphur is used in the vulcanization of rubber.

4. Synthetic Organic Chemistry

Sulphur containing compounds are widely employed in synthetic organic chemistry.

For example, Thiol (-SH) compounds are used in the synthesis of various organic molecules and pharmaceuticals.

5. Dyes and Pharmaceuticals

Sulphur compounds are used in the production of dyes, pigments, and pharmaceuticals.

6. Food Preservatives

Sulphur dioxide and sulphites (Na_2SO_3) are used as food preservatives.

This is because Sulphur dioxide and sulphites (Na_2SO_3) are reducing agents and inhibit the oxidation of food.

For More Information:

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